Extensive-form games

Construction of a behavior strategy from a mixed strategy

- This is slightly more involved.
- Recall that we have seen one example of a game where, for a given mixed strategy, there is no equivalent behavior strategy.
- However, that game did not have a perfect recall.

Example

- We now take a game with perfect recall for all the players.
- Consider the following game.

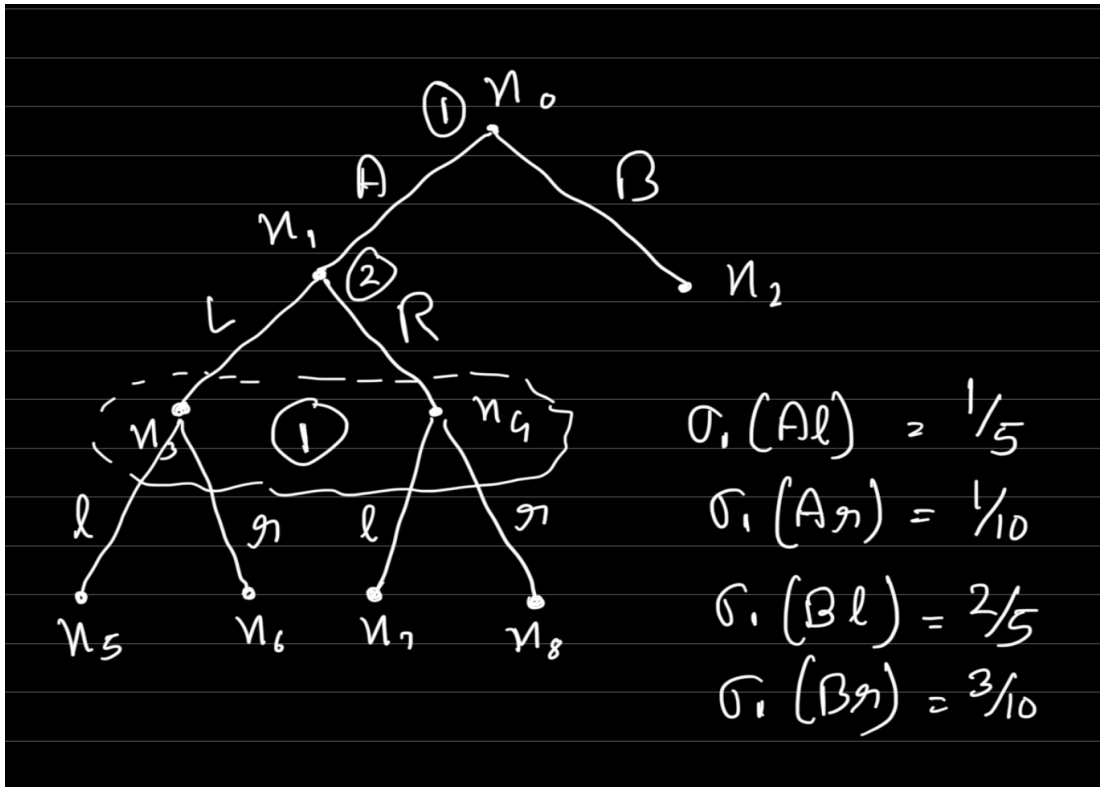

Figure: Figure 1

Example

- Suppose σ_1 is a mixed strategy of Player 1.
- Let h be the second information set of Player 1, and $a \in A(h)$.
- Let $S_1(h) \subseteq S_1$ denote the set of pure strategies of Player 1 such that the play of the game possibly reaches h, in other words, such that there exists a path through h containing the actions prescribed by the pure strategy under consideration.
- This means $S_1(h) = \{Al, Ar\}$. Further, let $\sigma_1(S_1(h))$ be the total probability assigned by σ_1 to the strategies in $S_1(h)$.

Example

- Within the set $S_1(h)$, consider those pure strategies that assign a to h, say $S_1(a; h)$.
- In the example, if we take $a = l$, then $S_1(l; h) = \{Al\}$, and if we take $a = r$, then $S_1(r; h) = \{Ar\}$.

Example

- Define $b_1(a; h) = \frac{\sigma_1(S_1(a;h))}{\sigma_1(S_1(h))}$ if $\sigma_1(S_1(h)) > 0$. Otherwise, define $b_1(a; h)$ arbitrarily.
- Note that this is the conditional probability that if the game reaches h, Player 1 plays the action a.
- In this case, as $\sigma_1(S_1(h)) > 0$, we have $b_1(l; h) = \frac{\sigma_1(S_1(l;h))}{\sigma_1(S_1(h))} = \frac{\sigma_1(Al)}{\sigma_1(Al)+\sigma_1(Ar)} = \frac{1/5}{1/5+1/10} = \frac{2}{3}$.

Example

- Similarly, we have

  $b_1(r;h) = \frac{\sigma_1(S_1(r;h))}{\sigma_1(S_1(h))} = \frac{\sigma_1(Ar)}{\sigma_1(Al)+\sigma_1(Ar)} = \frac{1/10}{1/5+1/10} = \frac{1}{3}.$
- And, $b_1(A;x_0) = \frac{\sigma_1(Al)+\sigma_1(Ar)}{\sigma_1(S_1)} = \frac{1}{5} + \frac{1}{10} = \frac{3}{10}.$
- $b_1(B;x_0) = \frac{\sigma_1(Bl)+\sigma_1(Br)}{\sigma_1(S_1)} = \frac{2}{5} + \frac{3}{10} = \frac{7}{10}.$

Result

Theorem (Kuhn (1957))

In every game in extensive form, if Player i has perfect recall, then for every mixed strategy of Player i, there exists an equivalent behavior strategy.

- Proof can be found in Solan's book, Page number 232.

Equilibria in behavior strategies

Theorem

If all the players in an extensive form game have perfect recall, then the game has a Nash equilibrium in behavior strategies.

Proof

- From the extensive form game, we can consider the corresponding strategic form game.
- Since the extensive form is finite, the corresponding strategic form game is finite. Hence, there exists an NE in mixed strategies, say $\sigma^* = (\sigma_i^*)_{i \in N}$.
- Moreover, as all the players have perfect recall, each σ_i^* has an equivalent b_i^*.
- Therefore, $u_i(\sigma^*) = u_i(b^*)$ for all $i \in N$ where $b^* = (b_1^*, \ldots, b_n^*)$.

Proof

- We claim that b^* is an NE.
- Consider another behavior strategy b_i of Player i.
- This means there exists σ_i such that σ_i and b_i are equivalent.
- Therefore, as σ^* is and NE, we have

  $u_i(\sigma^*) \geq u_i(\sigma_i, \sigma^*_{-i}) = u_i(b_i, b^*_{-i})$.
- Now as $u_i(\sigma^*) = u_i(b^*)$, b^* is an NE.

Finding behavior strategy equilibrium

- To find a behavior strategy equilibrium, one way is to find a mixed strategy equilibrium and then find the corresponding equivalent behavior strategies.
- Consider the following game:

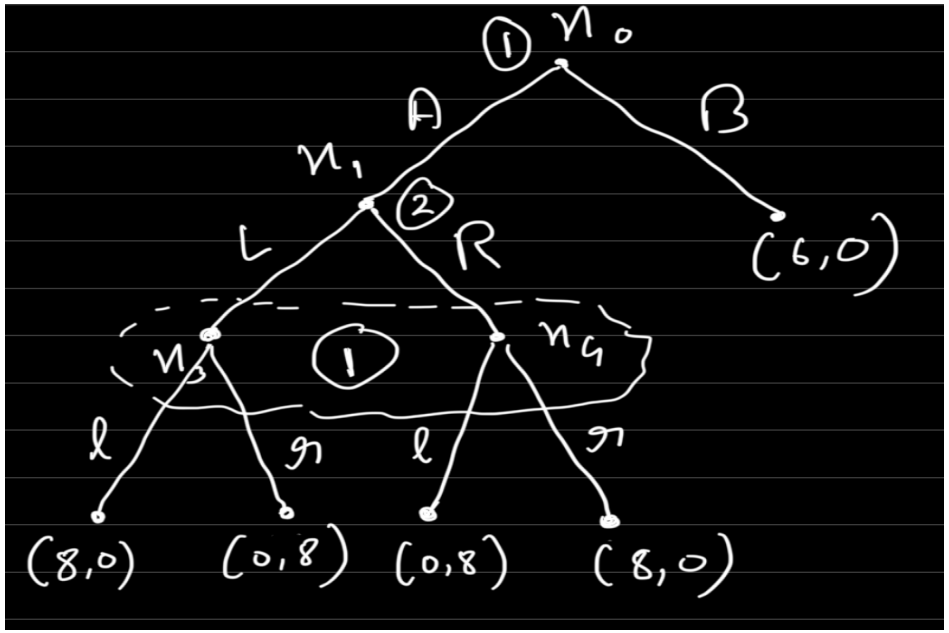

Figure: Figure 2

Finding behavior strategy equilibrium

- The corresponding strategic form game is

Table: Strategic form game

Player 1	Player 2	
	L	R
AC	(8,0)	(0,8)
AD	(0,8)	(8,0)
BC	(6,0)	(6,0)
BD	(6,0)	(6,0)

Finding behavior strategy equilibrium

- First observe that Player 2 will not play a pure strategy in any NE of the game.
- Let Player 2's strategy be $(\alpha[L], (1-\alpha)[R])$ where $0 < \alpha < 1$, and Player 1 play $(p_1[AC], p_2[AD], p_3[BC], p_4[BD])$.
- Since $0 < \alpha < 1$, Player 2 is indifferent between L and R, which implies that $p_1 = p_2$.

Finding behavior strategy equilibrium

- Suppose $p_1 = p_2 > 0$. Then we must have $8\alpha \geq 6$ and $8(1-\alpha) \geq 6$, which is impossible. Hence $p_3 + p_4 = 1$ and both $6 \geq 8\alpha$ and $6 \geq 8(1-\alpha)$, so $\frac{1}{4} \leq \alpha \leq \frac{3}{4}$.
- Therefore, any equilibrium in behavior strategy will be of the form $b_1(A) = 0$, $b_1(B) = 1$, $\frac{1}{4} \leq b_2(L) \leq \frac{3}{4}$. Finally, $b_1(C)$ may take any arbitrary value.

Subgames in games with imperfect information

- Recall the idea of subgames in perfect information games.
- The idea of a subgame is that it's a part of the original game that itself can be considered as a game.

Example

- Consider the following example of a game:

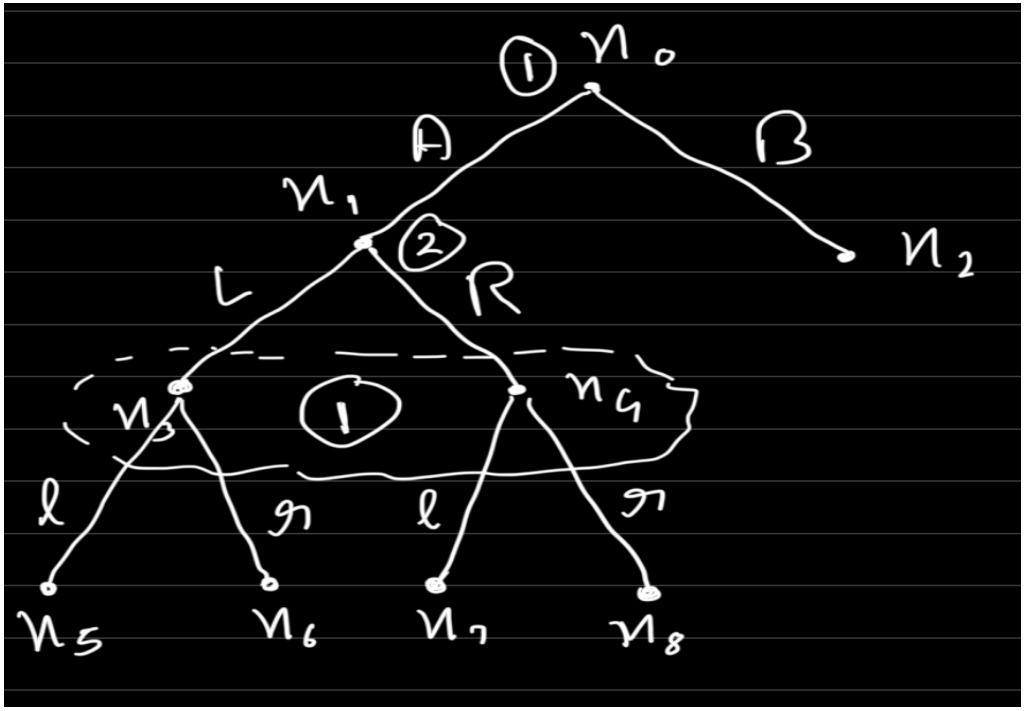

Figure: Figure 3

- Note that here there is one (non-trivial) subgame, starting at x_1. There are no subgames starting at x_3 or at x_4.

Definition

- We can define a subgame as in the case of perfect information game, but we must add this extra condition that every information set h of the original game Γ must either be contained in V^x or in $V \setminus V^x$.
- Note that this ensures for a subgame starting at a node x, x must be a singleton information set of some player in the game.

Subgame perfect equilibrium

- Similar to the case of perfect information game, a strategy profile σ (or b) is a subgame perfect equilibrium if for every subgame of Γ, the restriction of the strategy vector σ (or b) to that subgame is a Nash equilibrium in that subgame.

Subgame perfect equilibrium

- Note that unlike perfect information games, imperfect information games may not have subgame perfect equilibria in pure strategies.
- For example, consider the following game

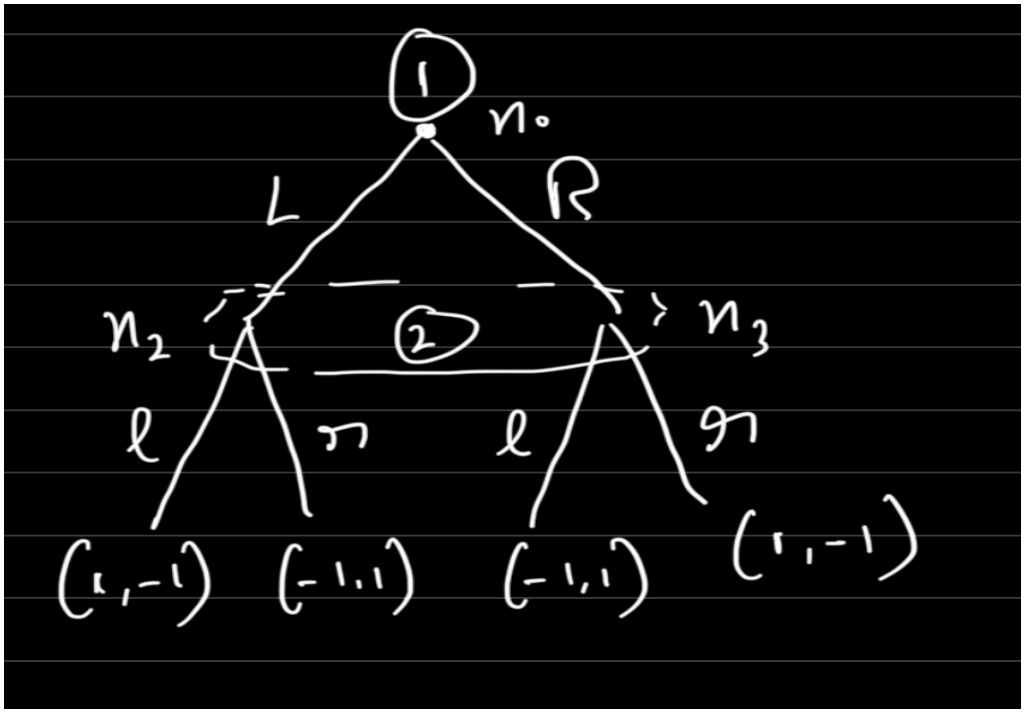

Figure: Figure 4

Backward induction

- The idea of backward induction can be used in imperfect information games as well.